Explainable AI: Opportunities and Limitations in Daily Bike-Rental Forecasting

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1. Introduction

This document summarizes the **advantages**, **limitations**, and **practical examples** of using Explainable Artificial Intelligence (XAI) in daily bike-rental forecasting.
The model outputs **numeric predictions** (estimated total daily rentals), while the **Data Provisioning Notebook** provides deeper visual explanations of which features influence those results.
Together, they ensure transparency and help users understand both *what* the model predicts and *why* those conditions matter.

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2. Opportunities (Advantages)

- **Transparency for decision-makers:**
  Daily forecasts show how weather and calendar factors shape demand, enabling more informed planning.

- **Improved stakeholder trust:**
  Planners can rely on interpretable numeric results, while the **Data Provisioning Notebook** offers detailed feature analysis for technical review.

- **Model debugging and validation:**
  The notebook verifies that the model behaves logically and uses meaningful weather-related inputs.

- **Regulatory and ethical alignment:**
  The project aligns with **EU Trustworthy AI** principles such as transparency, human oversight, and accountability.

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3. Limitations and Challenges

- **Scope limitation:**
  The model predicts only *daily totals* and cannot respond to hourly or real-time questions.

- **External factors:**
  Events like local festivals or competitions are not included, so the model may miss such exceptional days.

- **Interpretation boundaries:**
  The **Data Notebook** provides explanations and visualizations, while the model itself outputs only numeric results.

- **Seasonal assumptions:**
  If unusual weather occurs (for example, extreme rainfall in summer), predictions may still follow general seasonal patterns rather than precise anomalies.

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4. Practical Examples and Use Cases

(For detailed visualizations and explanations, users can refer to the Data Provisioning Notebook.)

- **Example 1:**
  Condition: Hot weather and very low wind speed.
  Model output: A high numeric value (e.g., 6,200 rentals) — favorable weather leads to high demand.

- **Example 2:**
  Condition: Cold winter day with a local cycling competition.
  Model output: Around 2,000 rentals — the model does not consider special events, so it cannot reflect unusual spikes in activity.

- **Example 3:**
  Condition: Requesting an hourly forecast (e.g., "How many rentals at 10 AM?").
  Model output: Not applicable — the model works at *daily granularity* and provides total daily counts only.

- **Example 4:**
  Condition: Summer day with high rainfall.
  Model output: Moderate number (e.g., 3,800 rentals) — rainfall lowers demand, but warm temperature keeps it from dropping too sharply.

- **Example 5:**
  Condition: Mild spring day with no rain and average wind speed.
  Model output: Above-average rentals (e.g., 5,100) — pleasant cycling conditions encourage consistent daily use.

These examples clarify how the AI responds under different scenarios and demonstrate both its **capabilities** and **boundaries** in real operational contexts.

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5. Balanced View

Explainable AI in this project supports **clarity, accountability, and operational trust**. By combining **daily numeric forecasts** with insights from the **Data Provisioning Notebook**, the system stays both understandable and reliable—ideal for planning and decision-making in real-world bike-rental management.